15 The Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET)

15.1 Motivation: Voltage-Controlled Switching and Amplification

The BJT is a powerful amplifier, but it has inherent limitations. The base current, while small, still flows into the device. This input current loads the preceding stage, consumes power, and limits input impedance. Furthermore, BJTs require a continuous base current to remain in the ON state, which is problematic for low-power digital logic where millions of transistors must switch efficiently.

What if we could build a three-terminal device where the input terminal draws **no DC current**? Such a device would have infinite input impedance, consume negligible input power, and be ideal for both high-impedance sensing and low-power digital switching. This is precisely what the **Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET)** provides.

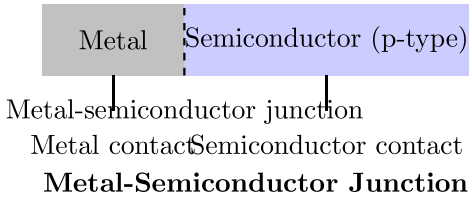
The Key Insight: Voltage Control, Not Current Control

The BJT is a **current-controlled current source**: $I_C = \beta I_B$.
The MOSFET is a **voltage-controlled current source**: $I_D = f(V_{GS})$.
This distinction is profound:

- In a BJT, the input terminal (base) draws current. You must supply power to control the device.
- In a MOSFET, the input terminal (gate) is electrically isolated by an insulating oxide layer. It draws **no DC current** (only a tiny transient current to charge the gate capacitance).
- This makes MOSFETs ideal for ultra-high-input-impedance amplifiers and for CMOS logic, where power is consumed only during switching.

15.2 The Metal-Semiconductor Junction: A Necessary Prelude

Before we can understand the MOSFET, we must understand the **metal-semiconductor junction**—the interface between a metal contact and the semiconductor that forms the gate terminal. This junction is fundamentally different from a PN junction.



Metal-Semiconductor Junction

15.2.1 Schottky Junction vs. Ohmic Contact

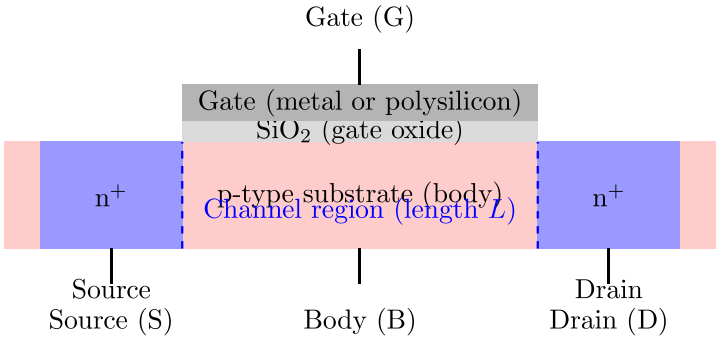
Depending on the metal work function and semiconductor doping, the metal-semiconductor junction can be one of two types:

- **Schottky junction (rectifying)**: Forms a potential barrier at the interface. Current flows easily in one direction (forward bias) but is strongly blocked in the opposite direction (reverse bias). This behaves like a diode, but with a lower forward voltage drop (≈ 0.3 V for silicon) than a PN junction (≈ 0.7 V).
- **Ohmic contact (non-rectifying)**: The barrier is so thin that carriers can tunnel through it regardless of bias direction. The junction exhibits linear current-voltage behavior: $I = V/R$. This is essential for making electrical connections to semiconductor devices without introducing unwanted diodes.

The MOS Structure: Metal-Oxide-Semiconductor

In a MOSFET, the gate terminal is a metal (or heavily doped polysilicon) layer separated from the semiconductor by a thin insulating layer of **silicon dioxide (SiO_2)**. This is not a metal-semiconductor junction—it is a **metal-insulator-semiconductor (MIS)** capacitor.
The key difference: The oxide layer is an **insulator**. No DC current can flow from the gate into the semiconductor. The gate voltage creates an **electric field** that penetrates through the oxide and modulates the carrier concentration in the semiconductor **without any gate current**.
This is the fundamental principle of the MOSFET: voltage control with zero DC input current.

15.3 MOSFET Structure and Operation



n-Channel MOSFET (nMOS) Structure

The MOSFET has four terminals:

- **Gate (G)**: Controls the transistor. Voltage applied here creates an electric field that modulates the channel conductivity.
- **Source (S)**: The terminal through which carriers enter the channel.
- **Drain (D)**: The terminal through which carriers exit the channel.
- **Body (B)**: The substrate (usually connected to the source for simplicity).

15.4 Physical Intuition: Channel Inversion

The term "inversion" captures the essence of MOSFET operation. In a p-type substrate, the majority carriers are holes. When a positive voltage is applied to the gate, the electric field penetrates the semiconductor and repels these holes away from the surface.

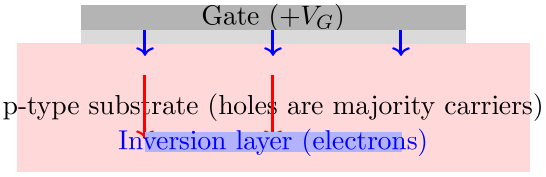


Figure: Channel Inversion - Electrons attracted to the surface

As holes are pushed away, a depletion region forms. When the gate voltage exceeds the threshold voltage V_{TH}, the electric field becomes strong enough to attract **minority carriers** (electrons) to the surface. The surface region **inverts** from p-type to n-type, creating a thin, continuous layer of mobile electrons. This layer is the **channel** that connects source to drain.

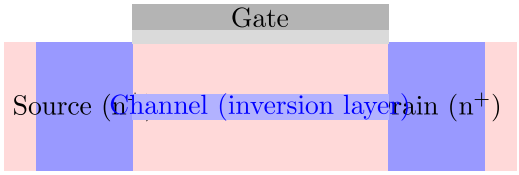


Figure: The channel connects source to drain when $V_{GS} > V_{TH}$

15.5 Physical Intuition: Pinch-Off and Current Saturation

Once the channel is formed, applying a drain voltage V_{DS} causes current to flow from drain to source. However, the channel is not uniform along its length.

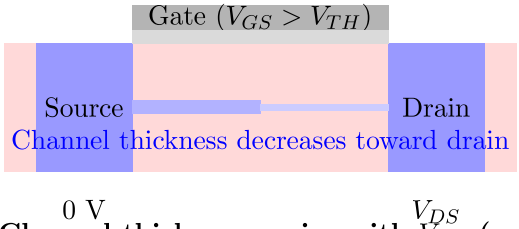


Figure: Channel thickness varies with V_{DS} (small V_{DS})

Why does the channel thin out toward the drain? The voltage along the channel increases from 0 V at the source to V_{DS} at the drain. The effective gate-to-channel voltage is $V_{GS} - V(x)$, where $V(x)$ is the voltage at position x. Near the drain, $V(x)$ is largest, so the local overdrive voltage $V_{GS} - V(x) - V_{TH}$ is smallest, resulting in a thinner inversion layer.

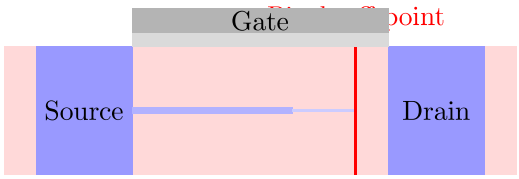


Figure: Pinch-off occurs when $V_{DS} = V_{GS} - V_{TH}$

When V_{DS} reaches $V_{GS} - V_{TH}$, the channel at the drain end is **pinched off**—the inversion charge goes to zero. For $V_{DS} > V_{GS} - V_{TH}$, the pinch-off point moves slightly toward the source, but the voltage drop across the remaining channel remains approximately constant at $V_{GS} - V_{TH}$. Electrons that reach the pinch-off point are swept into the drain by the strong electric field in the high-field region. The drain current saturates because further increases in V_{DS} do not increase the voltage drop across the channel.

15.6 MOSFET Operating Regions: A Practical Guide

Students often struggle to determine which formula to apply. Here is a simple decision tree:

How to Determine the MOSFET Operating Region

1. **Step 1: Check if the transistor is ON or OFF.**
   - If $V_{GS} < V_{TH}$: Transistor is in **cutoff**. $I_D = 0$.
   - If $V_{GS} \geq V_{TH}$: Transistor is ON. Proceed to Step 2.
2. **Step 2: Compare V_{DS} with $V_{GS} - V_{TH}$ (the overdrive voltage).**
   - If $V_{DS} < V_{GS} - V_{TH}$: Transistor is in **linear (triode) region**. Use:

$$I_D = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_{TH}) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

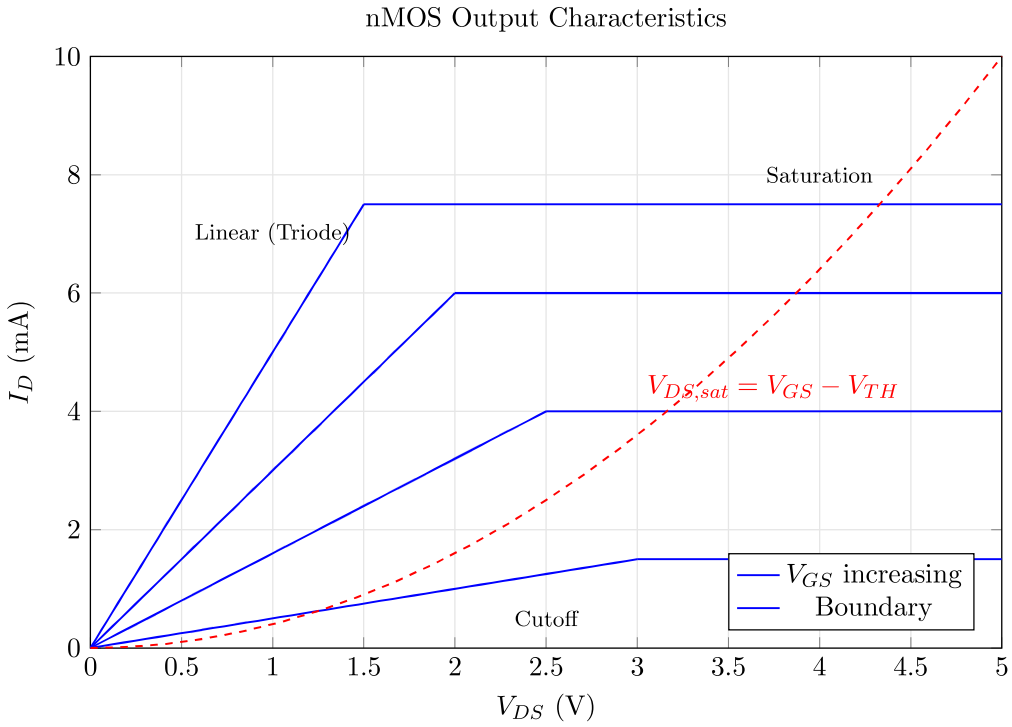
   - If $V_{DS} \geq V_{GS} - V_{TH}$: Transistor is in **saturation region**. Use:

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2$$

Quick Reference

$V_{GS} < V_{TH} \rightarrow$ Cutoff ($I_D = 0$)
$V_{GS} \geq V_{TH}$ and $V_{DS} < V_{GS} - V_{TH} \rightarrow$ Linear region
$V_{GS} \geq V_{TH}$ and $V_{DS} \geq V_{GS} - V_{TH} \rightarrow$ Saturation region

15.7 MOSFET Output Characteristics



15.8 Key MOSFET Equations Summary

MOSFET Equations (n-channel)

Cutoff region ($V_{GS} < V_{TH}$):

$$\boxed{I_D = 0}$$

Linear/Triode region ($V_{GS} \geq V_{TH}$, $V_{DS} < V_{GS} - V_{TH}$):

$$\boxed{I_D = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_{TH}) V_{DS} - \frac{V_{DS}^2}{2} \right]}$$

Saturation region ($V_{GS} \geq V_{TH}$, $V_{DS} \geq V_{GS} - V_{TH}$):

$$\boxed{I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2}$$

Transconductance (saturation):

$$\boxed{g_m = \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})}$$

15.9 The Threshold Voltage

The threshold voltage V_{TH} is the gate-to-body voltage at which the semiconductor surface **inverts** and a conducting channel forms.

$$\boxed{V_{TH} = V_{FB} + 2\phi_F + \frac{\sqrt{4\epsilon_s q N_a \phi_F}}{C_{ox}}}$$

where:

- V_{FB} is the flat-band voltage (accounts for metal-semiconductor work function difference and oxide charge)
- ϕ_F is the Fermi potential (distance from Fermi level to intrinsic level)
- N_a is the substrate doping concentration
- $C_{ox} = \epsilon_{ox}/t_{ox}$ is the gate oxide capacitance per unit area

For an n-channel MOSFET (nMOS), the threshold voltage is typically positive ($0.4 - 0.7$ V). For a p-channel MOSFET (pMOS), V_{TH} is negative.

15.9.1 Definition of Work Function

The **work function** ϕ of a material is the minimum energy required to remove an electron from the Fermi level to a point immediately outside the material (the vacuum level).

$$\boxed{\phi = E_{\text{vac}} - E_F}$$

- **Metals**: The work function is a well-defined material constant (e.g., aluminum: $\phi \approx 4.1$ eV, gold: $\phi \approx 5.1$ eV, polysilicon: $\phi \approx 4.6$ eV).
- **Semiconductors**: The work function is **doping-dependent** because the Fermi level position changes with doping.

The work function difference $\phi_{ms} = \phi_m - \phi_s$ contributes to the flat-band voltage V_{FB}.

15.10 CMOS: Complementary MOS for Digital Logic

The real revolution in digital electronics came from combining nMOS and pMOS transistors in **CMOS (Complementary MOS)** circuits. In a CMOS inverter:

- nMOS conducts when V_{GS} is high (logic 1)
- pMOS conducts when V_{GS} is low (logic 0)
- In steady state, one transistor is always OFF, so no DC current flows
- Power is consumed only during switching (dynamic power)

15.11 BJT vs. MOSFET: A Side-by-Side Comparison

Aspect	Bipolar Junction Transistor (BJT)	Metal-Oxide-Semiconductor FET (MOSFET)
Control mechanism	Current-controlled: I_B controls I_C	Voltage-controlled: V_{GS} controls I_D
Input impedance	Low (hundreds of Ω to kΩ)	Extremely high (virtually infinite DC input resistance)
Input current	I_B flows continuously	Zero DC gate current (only transient charging)
Output characteristic (active/saturation)	$I_C = \beta I_B$	$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{TH})^2$
Transconductance g_m	$g_m = \frac{I_C}{V_T}$	$g_m = \mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{TH})$
Switching speed	Fast, but suffers storage delay in saturation	Very fast; no minority carrier storage in channel
Power consumption	Quiescent power required (I_B flows)	Zero static gate current; only dynamic switching power
Scaling	Limited by base width and doping	Excellent scaling (Moore's Law for over 50 years)

Key Insight

The BJT and MOSFET are complements, not competitors:

- **BJTs** offer higher transconductance for the same bias current and are preferred in high-speed analog, RF, and low-noise applications.
- **MOSFETs** offer zero gate current, excellent scalability, and are the foundation of digital logic (CMOS) and low-power analog circuits.

15.12 Summary: The MOSFET as a Voltage-Controlled Switch and Amplifier

The MOSFET fundamentally changed electronics by offering a voltage-controlled switch with zero DC input current. This property enabled:

- Ultra-high-input-impedance amplifiers (ideal for sensors)
- CMOS digital logic with near-zero static power dissipation
- Continuous scaling (Moore's Law) for over five decades
- Integration of billions of transistors on a single chip

While BJTs remain superior for certain high-speed and high-current applications, the MOSFET is the undisputed workhorse of modern electronics, from microprocessors to memory chips to power management.